

The Cellular Level

The membrane decides what gets in. Almost everything else follows from that.

BY THE END OF THIS PAGE YOU SHOULD BE ABLE TO

- Describe the cell membrane and say why only some things cross it
- Tell passive transport from active transport, and give an example of each
- Predict what a cell does in a hypotonic, isotonic, or hypertonic solution
- Explain what the sodium–potassium pump does and why it costs ATP
- Say where ATP is made, and why a cell without it dies fast

1 A membrane that chooses

Every cell is wrapped in a double layer of fat with proteins floating in it.

That double layer is the **phospholipid bilayer**. The water-loving heads face out. The water-fearing tails tuck inside.

This makes a barrier. Small, greasy, uncharged things slip through easily. Oxygen, carbon dioxide, and steroid hormones all cross on their own.

Charged or large things cannot cross at all. So sodium, potassium, calcium, glucose, and amino acids each need **a protein to let them through**.

Those proteins are the control panel. When a drug, a hormone, or a disease changes physiology, it usually does it by changing one of these proteins.

THINK OF IT THIS WAY

The membrane is a nightclub wall, not a nightclub door. Nothing gets through the wall. Everything happens at the doors, and every door has its own bouncer.

2 Passive transport — downhill, and free

If something moves from where there is more of it to where there is less, the cell pays nothing.

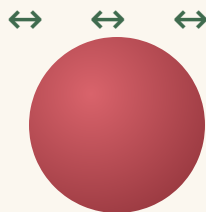
- **Simple diffusion.** Straight through the membrane. Oxygen and carbon dioxide do this all day. That is the basis of gas exchange.
- **Facilitated diffusion.** Still downhill, but through a protein. Glucose enters most cells this way.
- **Osmosis.** The special case. This is *water* moving toward the side with more dissolved solute.

One word matters here: **gradient**. Everything in passive transport runs down one. None of it needs ATP.

3 Osmosis and tonicity — what a solution does to a cell

Water follows solute. That one sentence answers most tonicity questions on an exam.

PUT A RED BLOOD CELL IN THREE SOLUTIONS



no net water movement — the cell holds its shape

SOLUTION	SOLUTE OUTSIDE	WATER MOVES	THE CELL
Hypotonic	Less than inside	Into the cell	Swells, and may burst
Isotonic	Same as inside	Both ways, equally	Stays the same
Hypertonic	More than inside	Out of the cell	Shrivels

WHY THE NURSE CARES

This is why IV fluids are chosen on purpose. Hang the wrong bag and you can swell cells that must not swell. Brain cells are the worst case, because there is no room for them to expand.

4 Active transport — uphill, and it costs

Moving something *against* its gradient always takes energy.

The main example is the **sodium-potassium pump**. For each ATP it burns, it pushes **3 sodium out** and pulls **2 potassium in**. Both moves are uphill.

- 1 **It keeps sodium low inside.** So sodium always has somewhere to rush when a channel opens.
- 2 **It keeps potassium high inside.** That is what sets the resting membrane potential.
- 3 **Three out, two in.** Each cycle removes one net positive charge, which helps make the inside negative.

About a fifth to a quarter of your resting energy goes to this one pump. That is the price of being ready to fire.

Bulk transport handles anything too big for a protein. **Endocytosis** brings material in by wrapping the membrane around it. Phagocytosis is the version immune cells use to eat bacteria. **Exocytosis** pushes material out. That is how neurons release neurotransmitter and glands release hormone.

5 Where the ATP comes from

Everything above runs on one kind of energy, and it is made in one place.

Glucose is broken down in the cytoplasm. It is finished off inside the **mitochondria**, using oxygen.

That is why oxygen delivery is what every emergency turns on. No oxygen means no ATP. No ATP means the pumps stop. When the pumps stop, sodium and water pour into the cell, and it swells and dies.

Cells with high energy needs are packed with mitochondria. Heart muscle, kidney tubules, and neurons are the big three. They are also the first to be injured when blood flow fails.

THINK OF IT THIS WAY

ATP is not stored in a cellar like fat. It is made minute to minute, like a kitchen cooking to order. Cut off the ingredients and service stops almost at once.

6 Check yourself

Five questions. The wrong answers explain themselves, so they are worth reading.

1. A red blood cell is put in a solution and starts to swell. What kind of solution is it?

- a) Hypertonic
- b) Isotonic
- c) Hypotonic
- d) Saturated with sodium

2. Which one of these costs the cell ATP?

- a) Oxygen diffusing into a cell
- b) The sodium-potassium pump
- c) Water moving by osmosis
- d) Glucose entering through a carrier, down its gradient

3. Why can oxygen cross the membrane on its own when sodium cannot?

- a) Oxygen is small and uncharged, and sodium is charged
- b) Oxygen is heavier than sodium
- c) Sodium is too big to fit anywhere
- d) Oxygen dissolves in water and sodium does not

4. A patient's cells lose their oxygen supply. Why do the cells swell?

- a) Oxygen normally pushes water out
- b) No ATP means the sodium pump stops, sodium builds up inside, and water follows
- c) The membrane dissolves without oxygen
- d) Carbon dioxide replaces the water inside

5. A macrophage wraps its membrane around a bacterium and swallows it. What is that called?

- a) Exocytosis
- b) Facilitated diffusion
- c) Phagocytosis, a type of endocytosis
- d) Active transport by a pump

Answer key

1. **Hypotonic**

Correct. Less solute outside means water moves in, and the cell swells.

2. **The sodium-potassium pump**

Right. It moves both ions uphill, and uphill always costs ATP.

3. **Oxygen is small and uncharged, and sodium is charged**

Exactly. The oily middle of the membrane blocks charged particles and lets small uncharged ones pass.

4. **No ATP means the sodium pump stops, sodium builds up inside, and water follows**

Correct. This same sequence is behind cell injury everywhere in the body.

5. **Phagocytosis, a type of endocytosis**

Right. This is bulk transport inward, and phagocytosis is the cell-eating version.

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